Waves - 1

→ Wave is a medium to transfer energy from one point to another.

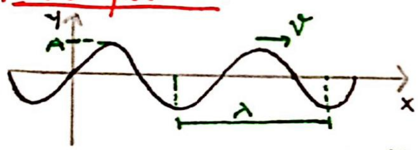
Transverse
Particle & wave move in ⊥ar directions
(String Waves)

Longitudinal
Particle & wave move in same direction
(Sound Waves)

Travelling Wave
Wave transferring energy from one point to another

Standing Wave
Wave formed due to superposition, that does not transfer energy.

#Wave Equation :



Initial Phase

$$\boxed{y = A\sin(\omega t - Kx + \phi)}$$

Ang. Frequency — Angular Wave No.

$$\boxed{\omega = 2\pi f = \frac{2\pi}{T}} \qquad \boxed{K = \frac{2\pi}{\lambda}}$$

Wave Speed ⇒

$$\boxed{v = \frac{\text{coeff. of } t}{\text{Coeff of } x} = \frac{\omega}{K} = f \cdot \lambda}$$

⊙ All the particles of a string wave execute SHM of same amplitude 'A'.

Phase & Path difference ⇒

$$\boxed{\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x}$$

Dirn of wave motion ⇒

(i) If ω & K have opp. sign ⇒ +ve x dirn
$y = A\sin(\omega t - Kx)$ / $A\sin(-\omega t + Kx)$

(ii) If ω & K have same sign ⇒ −ve x dirn
$y = A\sin(\omega t + Kx)$ / $A\sin(-\omega t - Kx)$

Particle velocity & Acceleration ⇒

Since, $y = A\sin(\omega t - Kx + \phi)$

$$\boxed{v_p = \frac{\partial y}{\partial t} = \omega\sqrt{A^2 - y^2}}$$

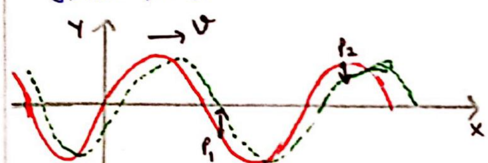
$$\boxed{a_p = \frac{\partial^2 y}{\partial t^2} = -\omega^2 y}$$

Particle direction at time t ⇒

1) $\boxed{v_p = -v \times \text{slope}}$

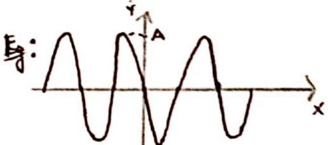
Here, v = wave velocity, with sign

2) Draw the future wave for very small time.



$P_1 \to$ Goes up
$P_2 \to$ Goes down

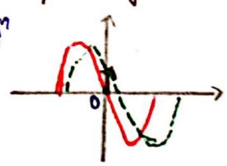
Eg:



$y = A\sin(\omega t + Kx + \phi)$

Snapshot of wave at t=0. Find ϕ.

Soln



Origin wala particle will move upward as wave moves ahead after t=0.

So, $v_p = \frac{\partial y}{\partial t} = -A\omega\cos(-\omega t + Kx + \phi)$

For origin particle at t=0, x=0

$v_p = -A\omega\cos(\phi)$

vp must be +ve as particle moves up, so,

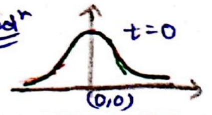
$\boxed{\phi = \pi}$ [As, $\cos\pi = -1$]

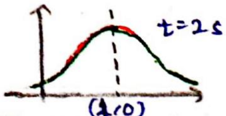
Eg: For a wave moving in +ve x,

$y = \frac{1}{1+x^2}$ at t=0 & $y = \frac{1}{1+(x-1)^2}$

at t=2s. Find wave velocity.

Soln





$v = \frac{1}{2} = 0.5\,m/s$ Ans

By graph transformation, graph at t=2sec was drawn.

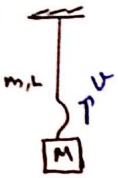
Wave speed on strings :

$$\boxed{v = \sqrt{\frac{T}{\mu}}}$$

Tension in string

Linear mass density of string (M/L)

⊙ If the mass of object suspended by string is comparitively very large, then assume tension to be constant.



(M >>> m)

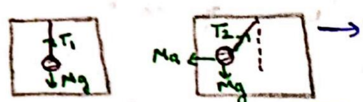
Here, $T \simeq Mg$

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg}{M/L}}$

$$\boxed{v = \sqrt{\frac{MgL}{m}}}$$

Eg: A ball (M) suspended from ceiling of car by string (m), also M >>> m. When car at rest, $v_{wave} = 60\,m/s$, when car at 'a' accn, $v_{wave} = 60.5\,m/s$. Find 'a'.

Soln



$T_1 = Mg$ $\qquad$ $T_2 = M\sqrt{a^2+g^2}$

Since μ of string is constant,

$v \propto \sqrt{T} \Rightarrow \frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}} = \left(\frac{M\sqrt{a^2+g^2}}{Mg}\right)^{1/2}$

So, $\frac{V_2}{V_1} = \left(1 + \frac{a^2}{g^2}\right)^{1/4} \Rightarrow \frac{60.5}{60} = \left(1 + \frac{a^2}{g^2}\right)^{1/4}$

$\Rightarrow \left(1 + \frac{0.5}{60}\right)^4 = 1 + \frac{a^2}{g^2}$

Binomial approx. $(1+x)^n = (1+nx)$

$\Rightarrow \cancel{1} + \frac{2}{30} = \cancel{1} + \frac{a^2}{g^2} \Rightarrow \boxed{a = \sqrt{\frac{g^2}{30}}}$ Ans

(*) When string has significant mass ⇒

Find time taken by pulse to reach top.

1) Tension

⊙ Tension at lowest pt is zero when free end.

⊙ Tension at each pt is diff.

⊙ Tension at a distance y from free end given by,

$$\boxed{T = \frac{M}{L} \cdot y \cdot g}$$

3) Acceleration of pulse ⇒

Since, $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mgy/L}{M/L}} = \sqrt{gy}$

[We calculate v & a of pulse for some pt. at y-distance.]

$$a = v.\frac{dv}{dy} = \sqrt{gy}\cdot\frac{\sqrt{g}}{2\sqrt{y}}$$

$\boxed{a = g/2}$ → Acc^n const. aaya hai, means newton's equation of motion are applicable.

3) Time to Reach top ⇒

By, $S = ut + \frac{1}{2}at^2$

$$L = (0)t + \frac{1}{2}\left(\frac{g}{2}\right)t^2$$

$$\Rightarrow \boxed{t = 2\sqrt{\frac{L}{g}}}$$

Avg. Power & Intensity :

$$\boxed{P_{avg} = \frac{1}{2}\rho S v \omega^2 A^2}$$

(Put, $\omega = 2\pi f$)

$$\boxed{P_{avg} = 2\pi^2 f^2 A^2 \rho v S}$$

Here, ρ = Density of string

S = Cross section of string

v = Wave speed = $\sqrt{T/\mu}$

$\omega = 2\pi f$

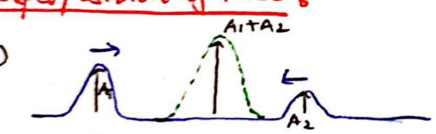
A = Amplitude

Also,

$$\boxed{I = \frac{P_{avg}}{S} = 2\pi^2 f^2 A^2 \rho v}$$

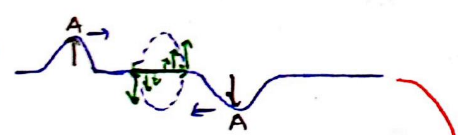
Superposition of Pulse :

(a)



(b)

(c)



[Complete destructive interference]

Imp

Although displacement is zero, but all energy is in form of pure KE

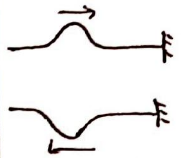
(Don't mistake it for PE)

Reflection from free & fixed end :

(a) When fixed end ⇒

↳ Pulse ulti hoke wapas aayegi i.e.

↳ Phase change of (π)

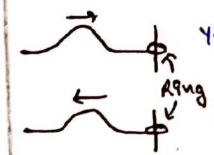


$y_i = A\sin(\omega t - kx)$

$y_r = A\sin(\omega t + kx + \pi)$

(b) When free End ⇒

↳ No phase change, only dir^n change

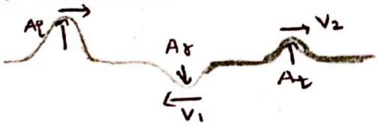


$y_i = A\sin(\omega t - kx)$

$y_r = A\sin(\omega t + kx)$

Reflection + Transmission b/w two strings :

(a)



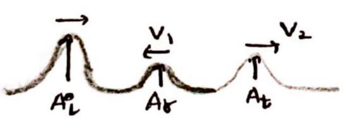
⊙ Strings of diff μ joined together (means same tension in both)

⊙ A pulse of A_i amp generated, some gets transmitted to thick string (A_t) rest reflected back (A_r).

Here, $v \propto \frac{1}{\sqrt{\mu}}$ [∴ T = const.]

⇒ $(V_2 < V_1)$ [∴ $\mu_2 > \mu_1$]

(b)



$(V_2 > V_1)$

Conclusion ⇒

1) Thick string behaves as fixed end reflection (π phase change)

2) Thin string behaves as free end reflection (no phase change)

And,

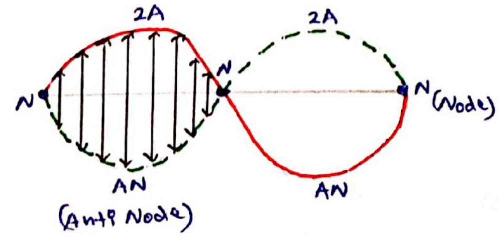
$$\boxed{A_t = \frac{2V_2}{V_1+V_2}\cdot A_i}$$

→ Sirf Numerator alag alag hai

$$\boxed{A_r = \frac{V_2-V_1}{V_1+V_2}\cdot A_i}$$

Standing / Stationary Wave :

→ This wave formed due to superposition of two Coherent travelling waves (i.e. same A, ω, f & λ) in opp. dir^n

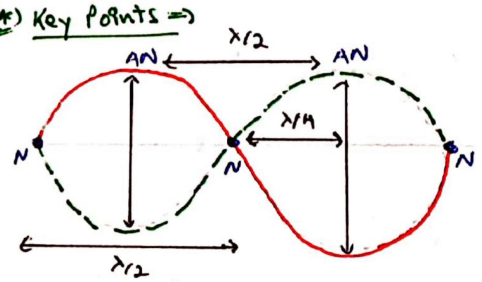


→ Amplitude of each pt. is diff.

→ Three blue pts will not have displacement at all, throughout wave & remain fix. They are (NODES)

→ Points with max Amplitude (2A) are (Anti-Nodes)

(*) Key Points ⇒



(i) All points b/w two successive nodes are in same phase

(ii) All points across a node (on left & right) are in opp. phase ($\Delta\phi = \pi$)

(iii) Distance b/w two consecutive Nodes or Anti-nodes is ($\lambda/2$)

(iv) Distance b/w consecutive node & anti-node is ($\lambda/4$)

Equation of Standing Wave :

$$\boxed{y = 2A\sin(kx+\phi_1)\sin(\omega t+\phi_2)}$$

This term only tells us about amp. of SHM of a point at x-distance

If $\phi_1 = 0$

$y = 2A\sin kx \cdot \sin(\omega t+\phi_2)$

$\boxed{\text{At } x=0,\ \text{NODE}}$

If $\phi_1 = \pi/2$

$y = 2A\cos kx \cdot \sin(\omega t+\phi_2)$

$\boxed{\text{At } x=0,\ \text{AN}}$

Waves - 2

Eg: If $y = 1.0\,mm \cos(1.57\,cm^{-1})x \sin(78.5\,s^{-1})t$.

Find distance of closest node to origin in $x > 0$ region.

Sol^n Eq^n mein cos tab hi aata hai jab origin pe AN banta hai.

→ And, distance b/w N & AN is $\lambda/4$

so, $K = 1.57 \Rightarrow \frac{2\pi}{\lambda} = 1.57$

$\frac{\lambda}{2} = \frac{\pi}{1.57} \Rightarrow \frac{\lambda}{4} = \frac{\pi}{3.14}$

i.e. $\boxed{\frac{\lambda}{4} = \text{distance} = 1\,cm}$ Ans

Standing Wave in clamped string :

→ Generally, standing waves are generated in a clamped string with the help of a tuning fork.

(I) Fixed at both ends ⇒

(a)

$f_1 = \frac{V}{\lambda}$

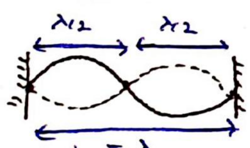
$\boxed{f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}}$

Because distance b/w 2 Nodes.

Least possible freq.

Called Fundamental freq. / 1st Harmonic

(b)



$f_2 = \frac{1}{L}\sqrt{\frac{T}{\mu}}$

$\Rightarrow \boxed{f_2 = \frac{2}{2L}\sqrt{\frac{T}{\mu}}}$

1st overtone & 2nd Harmonic

Hence, we can generalise,

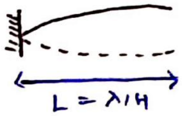
n^{th} Harmonic or $(n-1)^{th}$ overtone $\Rightarrow \boxed{f = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$ $\boxed{f = n \cdot f_1}$

⊙ If $f_1 = 100\,Hz$, then string will be in resonance with tuning fork of 100 Hz, 200 Hz, 300 Hz - - - .

Yaha jitne no. of AN honge, utne hi no. of loops banenge wave ke.

(II) Free at One End ⇒

(a)

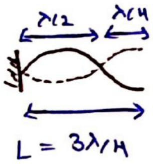


(Distance b/w N & AN)

$\boxed{f_1 = \frac{1}{4L}\sqrt{\frac{T}{\mu}}}$

Fundamental freq. or 1st Harmonic

(b)



$\boxed{f_3 = \frac{3}{4L}\sqrt{\frac{T}{\mu}}}$

3rd Harmonic
1st overtone

So,

$\boxed{f = \frac{2n+1}{4L}\sqrt{\frac{T}{\mu}} = (2n+1)f_1}$

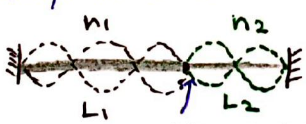
↳ $(2n+1)^{th}$ Harmonic

↳ n^{th} Overtone

⊙ If $f_1 = 100\,Hz$, then string will be in resonance with tuning fork of 100 Hz, 300 Hz, 500 Hz - - -

Standing Wave in Composite String :

→ Here frequency & Tension will always be same.



Node at junction.

Since,

$f_1 = f_2$

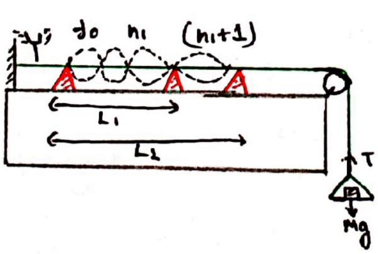
$\frac{n_1}{2L_1}\sqrt{\frac{T}{\mu_1}} = \frac{n_2}{2L_2}\sqrt{\frac{T}{\mu_2}}$

$\boxed{\frac{n_1}{n_2} = \frac{L_1}{L_2}\sqrt{\frac{\mu_1}{\mu_2}}}$

↳ used for ratio wale ques.

Sonometer :

→ Used to find wave ~~sp~~ speed in string.



→ We generate a standing wave of f_0 freq. by tuning fork & suppose we get a complete wave at L_1 distance.

→ So, then we move right wedge, just enough ahead so that we achieve one more loop of wave.

Hence,

$\boxed{L_2 - L_1 = \frac{\lambda}{2}}$

$\Rightarrow \lambda = 2(L_2 - L_1)$

So, $\boxed{v = f_0\lambda = 2f_0(L_2 - L_1)}$

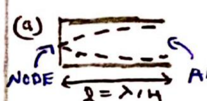
Organ Pipes :

(I) Closed Organ Pipes :→

→ One end of a cylindrical pipe is closed

→ Sound wave travels in it & reflects back, which superimpose & form stationary waves.

(a)



$\boxed{f_0 = \frac{v}{\lambda} = \frac{v}{4l}}$

Fundamental Freq. / 1st First Harmonic

Speed of sound wave

$\boxed{v = \sqrt{\frac{\gamma RT}{M}}}$ or $\boxed{v = \sqrt{\frac{\gamma P}{\rho}}}$

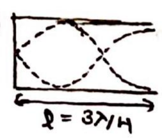
γ = Adiabatic const. = C_P/C_V

M = Molar Mass of Medium in pipe

P = Pressure in pipe

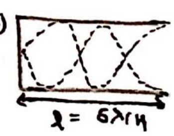
ρ = Density of med. in pipe.

(b)



$\boxed{f_3 = \frac{3v}{4l}}$ 3rd Harmonic / 1st overtone

(c)



$\boxed{f_5 = \frac{5v}{4l}}$ 5th Harmonic / 2nd Overtone

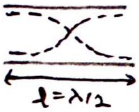
⊙ So, we can generalise

$\boxed{f_n = (2n+1)f_0}$ (For n^{th} overtone)

If frequency of tuning fork matches with odd multiple of fundamental freq, Resonance Occurs.

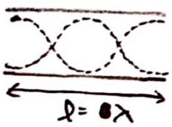
(II) Open Organ Pipe :→

(a)



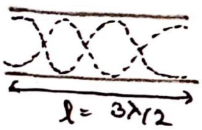
$\boxed{f_0 = \frac{v}{\lambda} = \frac{v}{2l}}$ 1st Harmonic / Fundamental Frequency

(b)



$\boxed{f_2 = \frac{2v}{2l}}$ 2nd Harmonic / 1st overtone

(c)



$\boxed{f_3 = \frac{3v}{2l}}$ 3rd Harmonic / 2nd overtone

So, we can generalise

$\boxed{f_n = (n+1) f_0}$ (For n^{th} overtone)

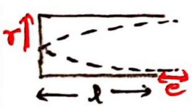
If freq. of Tuning Fork is integer multiple of fundamental freq. then Resonance Occurs.

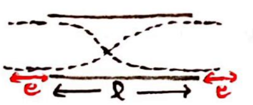
End Correction :→

→ In organ pipes, AN is formed a little outside the pipe.

→ This extra distance is end correction (e) & it depends of radius of pipe.

→ Hence, jaha bhi 'l' hai, uske saath 'e' ya '$2e$' add karna padega.





$\boxed{f_0 = \frac{v}{4(l+e)}}$ $\boxed{f_0 = \frac{v}{2(l+2e)}}$

$\boxed{e = 0.6r}$

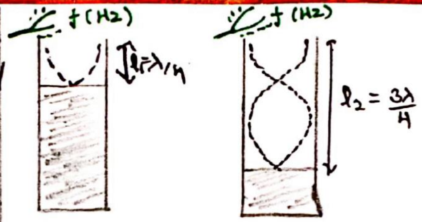
Resonance Tube :

→ It is an experiment, done to find 'v' of sound in air/medium.

→ Setup behaves like a closed organ pipe.

→ We create a standing wave with help of tuning fork whose f_0 & λ are known. (Means λ gets fixed for exp.)

→ Then we decrease water level to each overtone.



(i) Diff. b/w two consecutive Resonance / Harmonic is $\lambda/2$.

$$l_2 - l_1 = \lambda/2$$

$$\Rightarrow \boxed{\lambda = 2(l_2 - l_1)}$$

$$\Rightarrow \boxed{v = 2f(l_2 - l_1)}$$

(ii) End correction has no effect on final result & can be neglected.

Doppler's Effect :

→ When there is relative motion b/w source & observer, then frequency of sound recieved by observer is different than original frequency.

General case :→

$$\boxed{f_{app} = f_0\left(\frac{V \pm v_w \pm V_0}{V \pm v_w \pm V_s}\right)}$$

Here, V → speed of sound in Air

v_w → Velocity of wind

V_0 → speed of observer

V_s → speed of source.

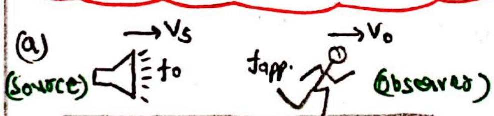
→ Decide plus-minus acc. to situation.

→ Put 'V_0' above & 'V_s' below.

(I) V_s & V_0 in same dir^n ⇒

If any speed tries to ↓ separation b/w source & observer, then sign is taken so as to ↑ f_{app} & vice-versa.

(a)



$V_0 = 0$	$V_s = 0$	Both moving
⇓	⇓	⇓
$f_0\left(\frac{V}{V-V_s}\right)$	$f_0 = \left(\frac{V-V_0}{V}\right)$	$f_0 = \left(\frac{V-V_0}{V-V_s}\right)$

(b)

Both moves, then

$$\boxed{f_{app} = f_0\left(\frac{V+V_0}{V+V_s}\right)}$$

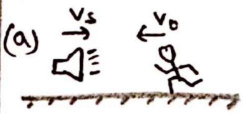
As, distance b/w obs. & sound is ↓, so f_{app} must ↑.

Num^r $(V+V_0)$ ↑ ⇒ f_{app} ↑

Den^r $(V+V_s)$ ↑ ⇒ f_{app} ↓

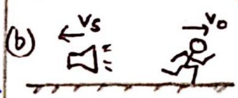
(II) V_s & V_0 in Opp. Dir^n :→

(a)



$\boxed{f_{app} = f_0\left(\frac{V+V_0}{V-V_s}\right)}$

(b)



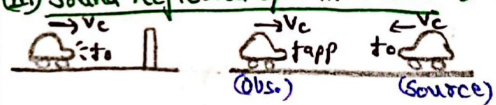
$\boxed{f_{app} = f_0\left(\frac{V-V_0}{V+V_s}\right)}$

Effect of Wind :→

If wind along v ⇒ $(v + v_w)$

If wind opp. v ⇒ $(v - v_w)$

(III) Sound Reflected by Fixed wall :→



1) Apply Doppler's effect formula 2 time

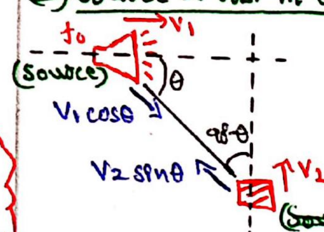
(OR)

2) Replace wall by mirror image of car & solve assuming original as observer

$$\boxed{f = f_0\left(\frac{V+V_c}{V-V_c}\right)}$$

$\boxed{\text{Beat frequency} = f_b = |f_{app} - f_0|}$

(IV) Source & obs. in diff. line :→



Take component of velocities along line joining them & solve.

$$\boxed{f_{app.} = f_0\left(\frac{V + V_2\sin\theta}{V - V_1\cos\theta}\right)}$$

Waves-3

Velocity of Sound Wave :

→ Since sound waves can travel through any medium solid, liq or gas.

$$\boxed{v = \sqrt{\frac{E}{\rho}}}$$

where, $\rho \to$ Density of med.

$E \to$ Modulus of elasticity

Y (solids) B (Liquid) $P / \gamma P$ (Gas)

→ In case of compressibility is given, we can calculate modulus of elasticity by

$$\boxed{K = \frac{1}{B} = \frac{1}{P} \Big/ \frac{1}{\gamma P}}$$

when Isothermal compression / when Adiabatic compression

→ Also, $P = \frac{\rho RT}{M}$, so we can write

$$\boxed{v = \sqrt{\frac{\gamma RT}{M}}}$$

(I) Variation of speed with temp :→

$\therefore v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow \boxed{v \propto \sqrt{T}}$

$\Rightarrow \boxed{\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}}$

⊙ If sound is travelling in a medium of distance varying temp, then

$t=0$ $t=t$ $\boxed{T = f(x)}$

We can calculate time to travel L length as,

$$v = K\sqrt{T}$$

$$\boxed{\int_0^L \frac{dx}{\sqrt{f(x)}} = \int_0^t K \cdot dt}$$

(II) Variation of speed with density :→

since, $P = \frac{\rho RT}{M}$

If we fix temp. (for a med.), then ratio of P & ρ becomes const.

i.e Velocity of sound is independent of pressure at a given temp.

[As, P/ρ = const. @ Fix temp]

→ But if we change med., then velocity will change.

Eg: $\boxed{v_{\text{Moist Air}} > v_{\text{Dry Air}}}$

Different forms of SHM Eqn :

(I) Imp differential eqn ⇒

⊙ $\boxed{\frac{\partial y}{\partial t} = -v \times \frac{\partial y}{\partial x}}$

⊙ $\boxed{\frac{\partial^2 y}{\partial t^2} = v^2 \times \frac{\partial^2 y}{\partial x^2}}$

[Also applicable For string waves]

(II) Displacement Eqn ⇒

$\boxed{y = A\sin(\omega t - Kx)}$

Put $\left(K = \frac{\omega}{v}\right)$ → $\boxed{y = A\sin\omega\left(t - x/v\right)}$

Put $\left(\omega = \frac{2\pi}{T}\right)$ → $\boxed{y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)}$

Take λ common → $\boxed{y = A\sin\frac{2\pi}{\lambda}(vt - x)}$

Spherical & Cylindrical Waves :

(I) Spherical Waves ⇒

→ Formed due to point source

→ At any point M, intensity given by

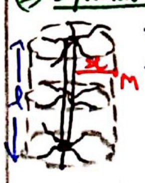
$\boxed{I_M = \frac{P}{4\pi x^2}} \Rightarrow \boxed{I \propto \frac{1}{x^2}}$

$[\therefore \sqrt{I} \propto A]$ $\boxed{A \propto \frac{1}{x}}$

→ Spherical wave eqn

$$\boxed{y = \frac{A}{x}\sin(\omega t - Kx)}$$

(II) Cylindrical Waves ⇒



→ Formed due to line source

→ At any point M, intensity given by

$$\boxed{I_M = \frac{P}{2\pi x l}}$$

Means, $\boxed{I \propto \frac{1}{x}}$ & $\boxed{A \propto \frac{1}{\sqrt{x}}}$

So, Cylindrical Wave eqn

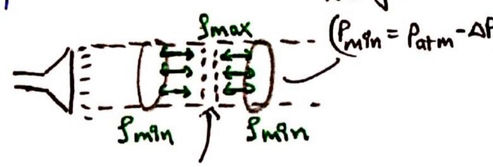
$$\boxed{y = \frac{A}{\sqrt{x}}\sin(\omega t - Kx)}$$

Sound Waves as Pressure Waves :

→ Sound waves travel by producing compressions & rarefactions, along its propogation.

→ Due to this, Pressure ↑ses at compression & ↓ses at Rarefaction.

→ This way, Sound behaves like a pressure wave.



For a sound wave

$$\boxed{y = A\sin(\omega t - Kx)}$$

$$\Rightarrow \boxed{\Delta P = \Delta P_0 \cos(\omega t - Kx)}$$

Pressure amplitude

$$\boxed{\Delta P_0 = \sqrt{2\rho v I} = \frac{2\pi}{\lambda}AB}$$

Here, I = Intensity of sound wave

A = Amplitude ~~(.....)~~ of sound

B = Bulk modulus of medium.

Imp. Point ⇒

(i) ΔP_0 can maximum become P_{atm}

(ii) If we put $\Delta P_0 = P_{atm}$, then we can find max. possible amp. of sound wave by $\Delta P_0 = \frac{2\pi}{\lambda}(AB)$.

Loudness Level :

We can measure loudness of sound wave of I intensity by

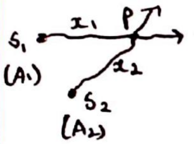
$$\boxed{L\ (\text{in dB}) = 10\log_{10}\left(\frac{I}{I_0}\right)}$$

I_0 = Reference Intensity = $10^{-12}\ \text{W/m}^2$

General Loudness Level ⇒

- 0 dB → silence
- 20-30 dB → Library volume
- 50-60 dB → Gen. Conversation
- 60-70 dB → Loud Talking
- 70-80 dB → Shouting
- 90-100 dB → Thunder
  ↳ (Painful for ears)

Interference of Waves :



→ Interference occurs only b/w coherent sources (same frequency & constant phase diff.)

→ After interference, resulting displacement at P is,

$$\boxed{y_P = R\sin(\omega t - \theta)}$$

$$\boxed{R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\phi_P)}}$$

$$\boxed{\tan\theta = \frac{A_2\sin(\phi_P)}{A_1 + A_2\cos(\phi_P)}}$$

$$\boxed{\text{Phase diff. at P} = \phi_P = \frac{2\pi}{\lambda}(x_2 - x_1)}$$

(*) Constructive Interference :→

→ R is max & $\cos\phi = +1$

$\boxed{\phi = 2n\pi}$ $(0, 2\pi, 4\pi, 6\pi ---)$

$\boxed{\Delta x = n\lambda}$ $(\lambda, 2\lambda, 3\lambda, ---)$

$\boxed{R_{max} = A_1 + A_2 = 2A}$

(*) Destructive Interference :→

→ R is min. & $\cos\phi = -1$

$\boxed{\phi = (2n\pm1)\pi}$ $(\pi, 3\pi, 5\pi --)$

$\boxed{\Delta x = (2n+1)\frac{\lambda}{2}}$ $(\lambda/2, 3\lambda/2, ---)$

$\boxed{R_{min.} = |A_1 - A_2| = 0}$

⇒ If $A_1 = A_2 = A$, then $\boxed{R = 2A\cos(\phi/2)}$

(*) Intensity at interference point :→

$$\boxed{I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cdot\cos\phi}$$

$\boxed{I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2 = 4I_0}$ Constructive condⁿ

$\boxed{I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2 = 0}$ Destructive condⁿ

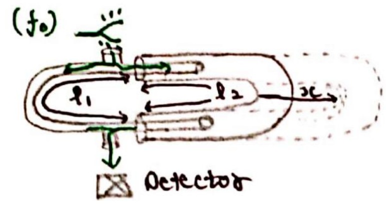
Also,

$$\boxed{\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I_1}+\sqrt{I_2})^2}{(\sqrt{I_1}-\sqrt{I_2})^2} = \left(\frac{A_1+A_2}{A_1-A_2}\right)^2}$$

Interference in Quincke's Tube :

→ Used to measure sound velocity in some ~~unkno~~ gaseous medium.

→ Two U-tubes of diff. radii. Larger radii U-tube, enveloped on smaller one such that silence is detected.



→ When tuning fork vibrated, sound travels in both dirⁿ & meet at detector to interfere destructively.

$$\boxed{\Delta x = l_2 - l_1 = (2n+1)\frac{\lambda}{2}} \quad -①$$

→ Now, Larger U-tube displaced by x-distance, such that again second time silence is detected.

$$\boxed{\Delta x = (l_2 + 2x) - l_1 = (2n+3)\frac{\lambda}{2}} \quad -②$$

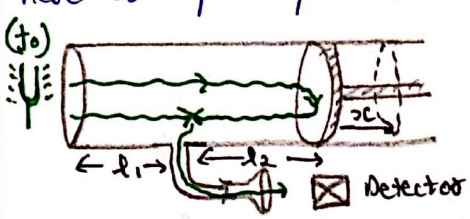
By ① & ② ⇒ $\boxed{\lambda = 2x}$

⇒ $\boxed{v = (2x)\times f_0}$

Interference in Seebeck's Tube :

→ This also used to measure sound velocity in some medium.

→ Same procedure as above, just difference is instead of larger U-tube, here we displace piston.



For first silence

$$\Delta = (l_1 + l_2 + l_2) - (l_1) = 2l_2 = (2n+1)\lambda/2$$

For second silence

$$\Delta = 2l_2 + x = (2n+3)\frac{\lambda}{2}$$

From above equations,

$\boxed{\lambda = 2x}$

$\boxed{v = 2f_0 x}$

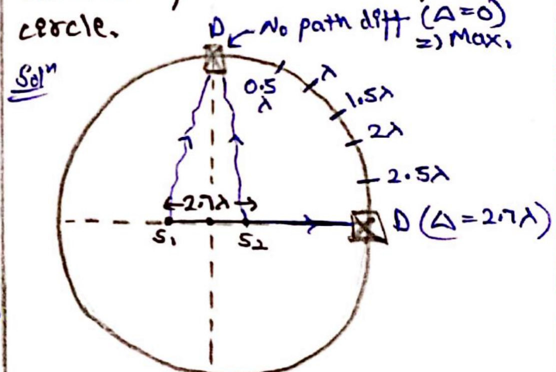
Note : (i) Above two experiments can also be performed for max. amplitude/constructive interference.

(ii) We perform for silence/destructive interference because its easier to detect silence over max. amp.

Interference of two closely placed sources in circular path :

Eg: Find total no. of max. & min. observed by 'D' in one complete circle.

Solⁿ



In one quarter

$\lambda, 2\lambda$ ⇒ 2 Max

$0.5\lambda, 1.5\lambda, 2.5\lambda$ ⇒ 3 Min.

At four diametrical ends

$\Delta = 0$

⇒ Max. at each end

So ; total Max = 10
Min = 12

Tuning Fork :

1) $f_0 \downarrow$ if l is increased
$f_0 \uparrow$ if l is decreased

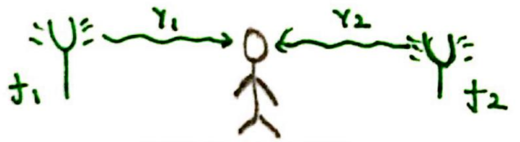
2) $f_0 \downarrow$ if prongs are loaded (by wax)
$f_0 \uparrow$ if prongs are unloaded/rubbed by sand paper.

Waves Remaing Part

Beats :

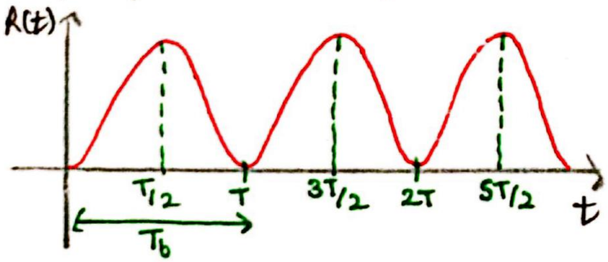
→ Beats are produced when waves (sound) of different frequency are superposed.

→ The resultant amplitude of resulting beat increases & decreases continuously with time.



$$\boxed{y_R = y_1 + y_2}$$

Graph of Resultant amplitude



$$\boxed{\text{Beat freq.} = f_b = |f_2 - f_1|}$$

$$\boxed{\text{Beat period} = T_b = \frac{1}{f_b}}$$

No. of beats occuring per second.

→ Human ears can only detect beat freq. of 10Hz–12Hz.

→ Beyond this range, sound appears continuous only to our ears.